

Jiyeol Oh

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RESEARCH INTERESTS

Vehicle Dynamics and Control

- Autonomous driving: Risk assessment, Path planning, Path tracking
- Integrated chassis control: Anti-lock Braking System, Electronic Stability Control, Torque Vectoring
- Parameter & state estimation
- Steady-state analysis: Handling diagram, Region of Attraction (RoA) estimation

Control Theory

- Nonlinear stability analysis: Bifurcation theory
- Modal analysis: Mode classification and tracking, Modal controllability and observability
- Robust and adaptive control: Linear Matrix Inequalities, H_∞ control, Disturbance observer

Vehicle-trailer

- Axle load optimization
- Weight distribution hitch
- Trailer sway & swing detection and mitigation

EDUCATION

Ph.D. in Mechanical Engineering, KAIST

Mar 2022 – Aug 2026 (expected)

- **Dissertation:** Open-Loop System Optimization of Light Vehicle-Trailer Combinations for Improving Stability and Maneuverability
- **Advisor:** Prof. Seibum Choi

M.S. in Mechanical Engineering, KAIST

Mar 2020 – Feb 2022

- **Thesis:** Torque Distribution Strategy of a In-Wheel-Motored Vehicle based on Tire-Vehicle Interactive Model
- **Advisor:** Prof. Seibum Choi

B.S. in Mechanical Engineering, KAIST

Mar 2015 – Feb 2020

PUBLICATIONS

[J1] **Jiyeol Oh** and Seibum Choi, "Optimal tongue weight selection criteria for light vehicle-trailer combinations," *Vehicle System Dynamics*, 63(12), 2347–2370, 2025.

[J2] **Jiyeol Oh** and Seibum Choi, "Operating principles of weight distribution hitches and their impact on handling characteristics of vehicle-trailer systems," *Vehicle System Dynamics*, 1–27, 2026. (Early access)

[C1] Semin Jeong, **Jiyeol Oh**, and Seibum Choi, "Design of a Sensorless Trailer Anti-Lock Braking System Based on Trailer Swing Detection," *2026 American Control Conference (ACC)*, New Orleans, LA, USA, 2026.

[C2] **Jiyeol Oh**, Haewoo Lee, and Seibum Choi, "Lateral Dynamics Modeling of General n -unit Combination Vehicles via Constraint Augmentation," *2026 IEEE Conference on Decision and Control (CDC)*, Honolulu, Hawaii, USA, 2026. (Accepted)

[J3] Haewoo Lee, **Jiyeol Oh**, and Seibum Choi, "Efficient Collision Avoidance Trajectory Optimization for Autonomous Vehicles based on Multiple Time Constants Point Mass Model," *Vehicle System Dynamics*. (Under review)

[C3] **Jiyeol Oh**, Haewoo Lee, and Seibum Choi, "Modal Family Classification for Vehicle-Trailer-Driver Lateral Dynamics," *2026 International Conference on Control, Automation and Systems (ICCAS)*, Sapporo, Japan, 2026. (Under review)

[J4] **Jiyeol Oh**, Haewoo Lee, Jinhwan Jeon, and Seibum Choi, "Modal Analysis Framework for Vehicle-Trailer Lateral Dynamics: Mode Classification, Tracking, and Analysis," *Vehicle System Dynamics*. (In progress)

RESEARCH PROJECTS

Development of Control Algorithm for Enhanced Stability in Tow-Mode of xEV

Mar 2024 – Jul 2026

- **PI:** Prof. Seibum Choi
- **Sponsor:** Hyundai Motor Company
- **Role:** Project Lead
- **Contributions:** Parameter & state estimation; vehicle-trailer stability index development; trailer sway and swing detection and mitigation

Development of Lv.4+ Autonomous Driving Vehicle Platform for Point-to-Point Driving to Logistics Centers for Heavy Trucks

Apr 2022 – Present

- **PI:** Prof. Seibum Choi
- **Sponsor:** Ministry of Trade, Industry and Energy (MOTIE)
- **Role:** Project Lead
- **Contributions:** Parameter estimation; fault diagnosis and fault-tolerant control; Minimal Risk Maneuver (MRM) strategy development

Driver-Customized Acceleration/Deceleration Pattern Derivation

Sep 2020 – Aug 2021

- **PI:** Prof. Seibum Choi
- **Sponsor:** Hyundai Motor Company
- **Role:** Research Assistant
- **Contributions:** Driver-specific feature extraction based on Discrete Wavelet Transform (DWT)

SKILLS

Software : MATLAB, Simulink, CarSim

Programming : Python

Languages : Korean (Native), English (Business), Japanese (Intermediate)